Credit Risk - Types and Factors

What is Credit Risk?

Credit risk is the probability of a financial loss resulting from a borrower's failure to repay a loan. Essentially, credit risk refers to the risk that a lender may not receive the owed principal and interest, which results in an interruption of cash flows and increased costs for collection. Lenders can mitigate credit risk by analysing factors about a borrower's creditworthiness, such as their current debt load and income.

Credit risks are calculated based on the borrower's overall ability to repay a loan according to its original terms. To assess credit risk on a consumer loan, lenders often look at the five Cs of credit: credit history, capacity to repay, capital, the loan's conditions, and associated collateral.

Some companies have established departments responsible for assessing the credit risks of their current and potential customers. Technology has allowed businesses to quickly analyze data used to determine a customer's risk profile.

Bond credit-rating agencies, such as Moody's Investors Services and Fitch Ratings, evaluate the credit risks of corporate bond issuers and municipalities and then rate them. If an investor considers buying a bond, they will often review the credit rating of the bond. If a bond has a low rating (< BBB), the issuer has a relatively high risk of default. Conversely, if it has a stronger rating (BBB, A, AA, or AAA), the risk of default is lower.

a mortgage applicant with a superior credit rating and steady income is likely to be perceived as a low credit risk, so they will likely receive a low interest rate on their mortgage. In contrast, an applicant with a poor credit history may have to work with a subprime lender to get financing.

The best way for a high-risk borrower to get lower interest rates is to improve their credit score. If you have poor credit, consider working with a credit repair company.

Similarly, bond issuers with less-than-perfect ratings offer higher interest rates than those with perfect credit ratings. The issuers with lower credit ratings use high returns to entice investors to assume the risk associated with their offerings.

Types of Credit Risk

The following are the main types of credit risks:

1. Credit default risk

Credit default risk occurs when the borrower is unable to pay the loan obligation in full or when the borrower is already 90 days past the due date of the loan repayment. The credit default risk may affect all credit-sensitive financial transactions such as loans, bonds, securities, and derivatives.

The level of default risk can change due to a broader economic change. It can also be due because of a change in a borrower's financial situation, such as increased competition or recession, which can affect the company's ability to set aside principal and interest payments on the loan.

2. Concentration risk

Concentration risk is the level of risk that arises from exposure to a **single counterparty or sector,** and it offers the potential to produce large amounts of losses that may threaten the lender's core operations. The risk results from the observation that more concentrated portfolios lack diversification, and therefore, the returns on the underlying assets are more correlated.

For example, a corporate borrower who relies on one major buyer for its main products has a high level of concentration risk and has the potential to incur a large amount of losses if the main buyer stops buying their products.

3. Country risk

Country risk is the risk that occurs when a country **freezes foreign currency payment obligations**, resulting in a default on its obligations. The risk is associated with the country's political instability and macroeconomic performance, which may adversely affect the value of its assets or operating profits. The changes in the business environment will affect all companies operating within a particular country.

Factors Affecting Credit Risk Modeling

In order to minimize the level of credit risk, lenders should forecast credit risk with greater accuracy. Listed below are some of the factors that lenders should consider when assessing the level of credit risk:

1. Probability of Default (POD)

The probability of default, sometimes abbreviated as POD, is the likelihood that a borrower will default on their loan obligations. For individual borrowers, POD is based on a combination of two factors, i.e., credit score and debt-to-income ratio.

The POD for corporate borrowers is obtained from credit rating agencies. If the lender determines that a potential borrower demonstrates a lower probability of default, the loan will come with a low interest rate and low or no down payment on the loan. The risk is partly managed by pledging collateral against the loan.

2. Loss Given Default (LGD)

Loss given default (LGD) refers to the amount of loss that a lender will suffer in case a borrower defaults on the loan.
For example, assume that two borrowers, A and B, with the same debt-to-income ratio and an identical credit score. Borrower A takes a loan of $10,000 while B takes a loan of $200,000. The two borrowers present with different credit profiles, and the lender stands to suffer a greater loss when Borrower B defaults since the latter owes a larger amount. Although there is no standard practice of calculating LGD, lenders consider an entire portfolio of loans to determine the total exposure to loss.

3. Exposure at Default (EAD)

Exposure at Default (EAD) evaluates the amount of loss exposure that a lender is exposed to at any particular time, and it is an indicator of the risk appetite of the lender. EAD is an important concept that references both individual and corporate borrowers. It is calculated by multiplying each loan obligation by a specific percentage that is adjusted based on the particulars of the loan.

Risk Types

Risk Type	Definition	Key Examples	Impact
Credit Default Risk	The risk that a borrower fails to repay the loan or meet contractual obligations.	A customer defaults on their loan payment or misses credit card dues.	Leads to financial losses for the lender; requires provisioning for bad debts.
Concentration Risk	The risk arising from over-reliance on a small number of borrowers, sectors, or regions.	A bank lends most of its funds to a single industry like real estate or a few large corporations.	Losses increase significantly if the specific sector or borrower group faces challenges.

Country Risk	The risk of loss due to economic, political, or social instability in a specific country.	A company invests heavily in a country that later experiences a currency crisis or political turmoil.	Impacts business operations, reduces returns, and may cause asset value to decline in that country.